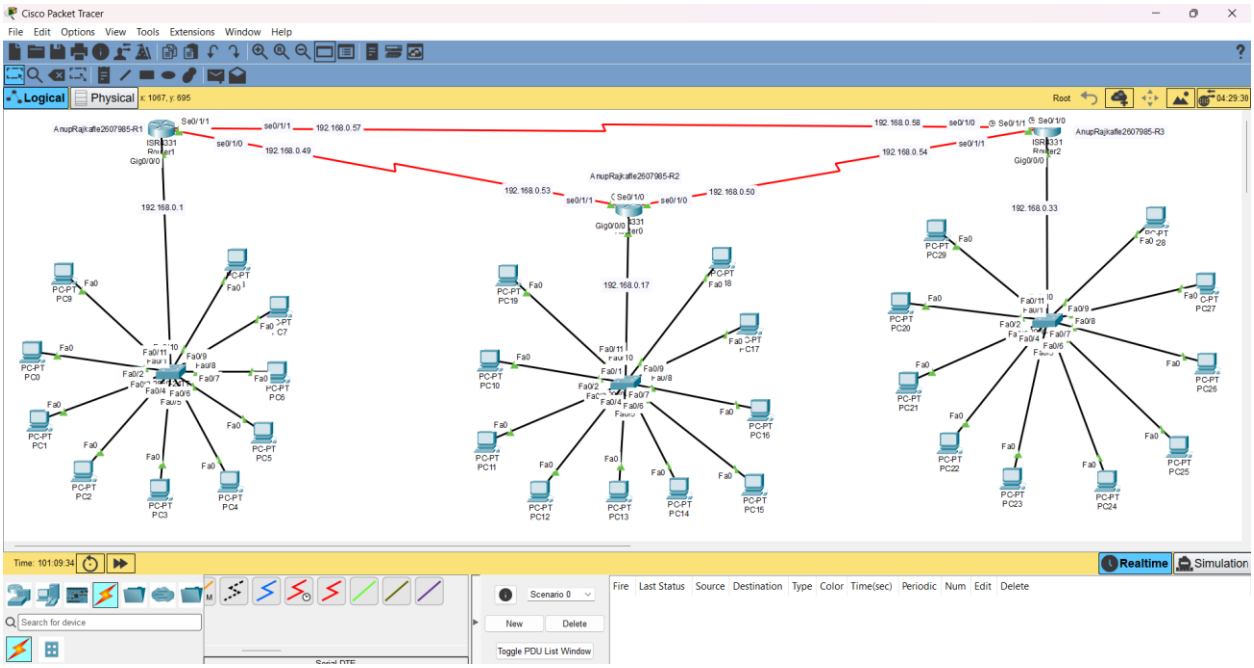


IP Planning

Subnet Purpose	Network Address	Subnet Mask	Wildcard Mask	Usable Host Range	Interface IPs / Gateway
AnupRajkafle 2607985-R1 LAN	192.168.0.0/28	255.255.255.240	0.0.0.15	192.168.0.1 – 192.168.0.14	Gateway (Gig0/0/0): 192.168.0.1
AnupRajkafle 2607985-R2 LAN	192.168.0.16/28	255.255.255.240	0.0.0.15	192.168.0.17 – 192.168.0.30	Gateway (Gig0/0/0): 192.168.0.17
AnupRajkafle 2607985-R3 LAN	192.168.0.32/28	255.255.255.240	0.0.0.15	192.168.0.33 – 192.168.0.46	Gateway (Gig0/0/0): 192.168.0.33

Subnet Purpose	Network Address	Subnet Mask	Wildcard Mask	Usable Host Range	Interface IPs / Gateway
AnupRajkafle2607985-R1 ↔ AnupRajkafle2607985-R2 Link	192.168.0.48/30	255.255.255.252	0.0.0.3	192.168.0.49 – 192.168.0.50	AnupRajkafle2607985-R1 Se0/1/0: .49 | AnupRajkafle2607985-R2 Se0/1/1: .50
AnupRajkafle2607985-R2 ↔ AnupRajkafle2607985-R3 Link	192.168.0.52/30	255.255.255.252	0.0.0.3	192.168.0.53 – 192.168.0.54	AnupRajkafle2607985-R2 Se0/1/0: .53 | AnupRajkafle2607985-R3 Se0/1/1: .54
AnupRajkafle2607985-R1 ↔ AnupRajkafle2607985-R3 Link	192.168.0.56/30	255.255.255.252	0.0.0.3	192.168.0.57 – 192.168.0.58	AnupRajkafle2607985-R1 Se0/1/1: .57 | AnupRajkafle2607985-R3 Se0/1/0: .58

Router running configuration

```
AnupRajkafle2607985-R1#show run
Building configuration...

Current configuration : 1045 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname AnupRajkafle2607985-R1
!
!
!
!
ip dhcp excluded-address 192.168.0.1
!
ip dhcp pool lan1_pool
 network 192.168.0.0 255.255.255.240
 default-router 192.168.0.1
!
!
!
no ip cef
no ipv6 cef
```

OSPF neighbor table

```
AnupRajkafle2607985-R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:37    192.168.0.50    Serial0/1/0
3.3.3.3           0   FULL/  -        00:00:32    192.168.0.58    Serial0/1/1
AnupRajkafle2607985-R1#
```

OSPF routing table

```
AnupRajkafle2607985-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.0.0/24 is variably subnetted, 9 subnets, 3 masks
C       192.168.0.0/28 is directly connected, GigabitEthernet0/0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0/0
O       192.168.0.16/28 [110/65] via 192.168.0.50, 00:40:19, Serial0/1/0
O       192.168.0.32/28 [110/65] via 192.168.0.58, 01:33:00, Serial0/1/1
C       192.168.0.48/30 is directly connected, Serial0/1/0
L       192.168.0.49/32 is directly connected, Serial0/1/0
O       192.168.0.52/30 [110/128] via 192.168.0.50, 00:40:19, Serial0/1/0
                        [110/128] via 192.168.0.58, 00:40:19, Serial0/1/1
C       192.168.0.56/30 is directly connected, Serial0/1/1
L       192.168.0.57/32 is directly connected, Serial0/1/1
```

Successful ping tests

```
AnupRajkafle2607985-R1#ping 192.168.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/11/45 ms

AnupRajkafle2607985-R1#ping 192.168.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/6 ms

AnupRajkafle2607985-R1#ping 192.168.0.33

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.33, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 25/26/29 ms

AnupRajkafle2607985-R1#ping 192.168.0.54

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.54, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 25/29/36 ms
```

A short paragraph explaining: o How you chose subnet masks and wildcard masks, o How you configured OSPF

In order to make sure that IP addresses are distributed properly, the technology known as Variable Length Subnet Masking (VLSM) is introduced. With respect to the calculation of subnet block size for LANs that will have 10 hosts, the block size of 16 ($2^4 =16$) is chosen which specifies a mask of /28 (255.255.255.240). Point to point links only need 2 host IPs hence block size of 4 has been chosen ($2^2 = 4$), which results in a mask of /30 (255.255.255.252). To achieve this thing, wildcard masks were successfully calculated by subtracting each subnet mask from (255.255.255.255). The whole process yielded 0.0.0.15 for LAN subnets and 0.0.0.3 for WAN links. The OSPF dynamic routing was used in all three routers using ID 1. Each router was assigned a unique router ID (1.1.1.1, 2.2.2.2, and 3.3.3.3) and shared subnet numbers associated to their connections through OSPF.